

3. 电路如图 4.3.3 所示, 则 a、b 端的戴维南等效电路和诺顿等效电路中的 $U_{oc} =$

3V, $Z_0 =$ 3Ω , $I_{sc} =$ 1A。

4. 电路图 4.3.4 所示, 已知 $X_L = 100\Omega$, $X_C = 200\Omega$, $R = 150\Omega$, $U_C = 100V$, 则电压 $U =$ 117.85V, 电流 $I =$ ~~$5/2$~~ $5/6$ A。 kVL

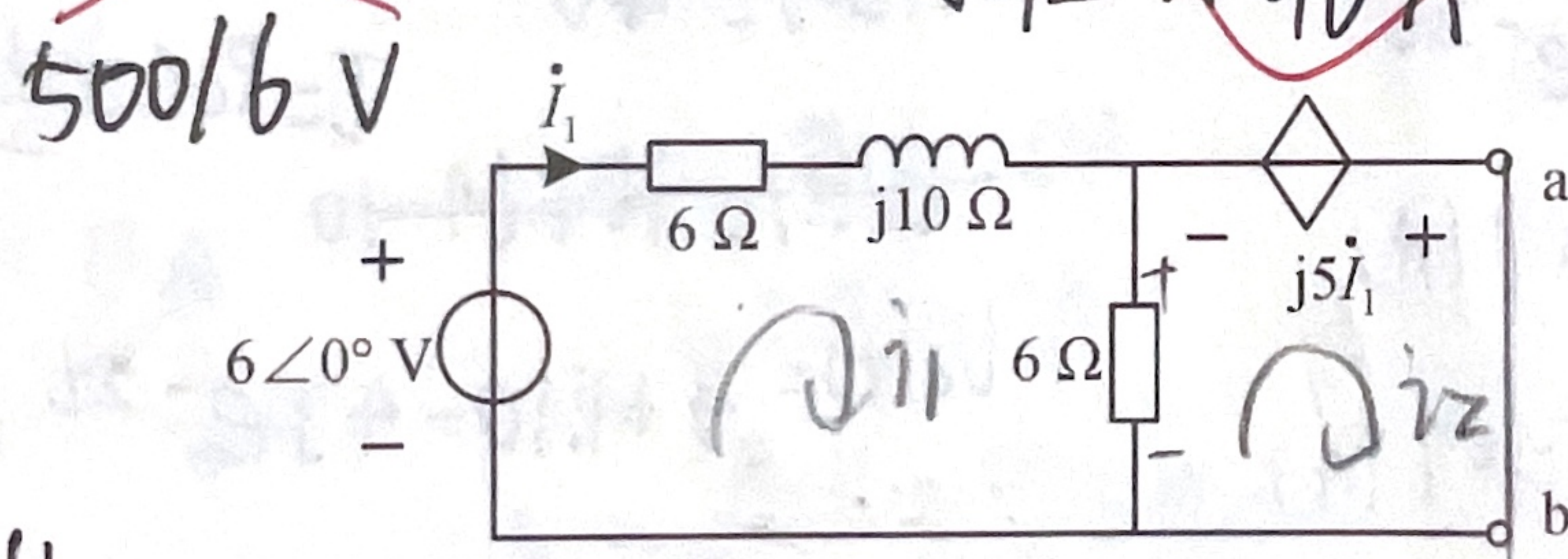


图 4.3.3

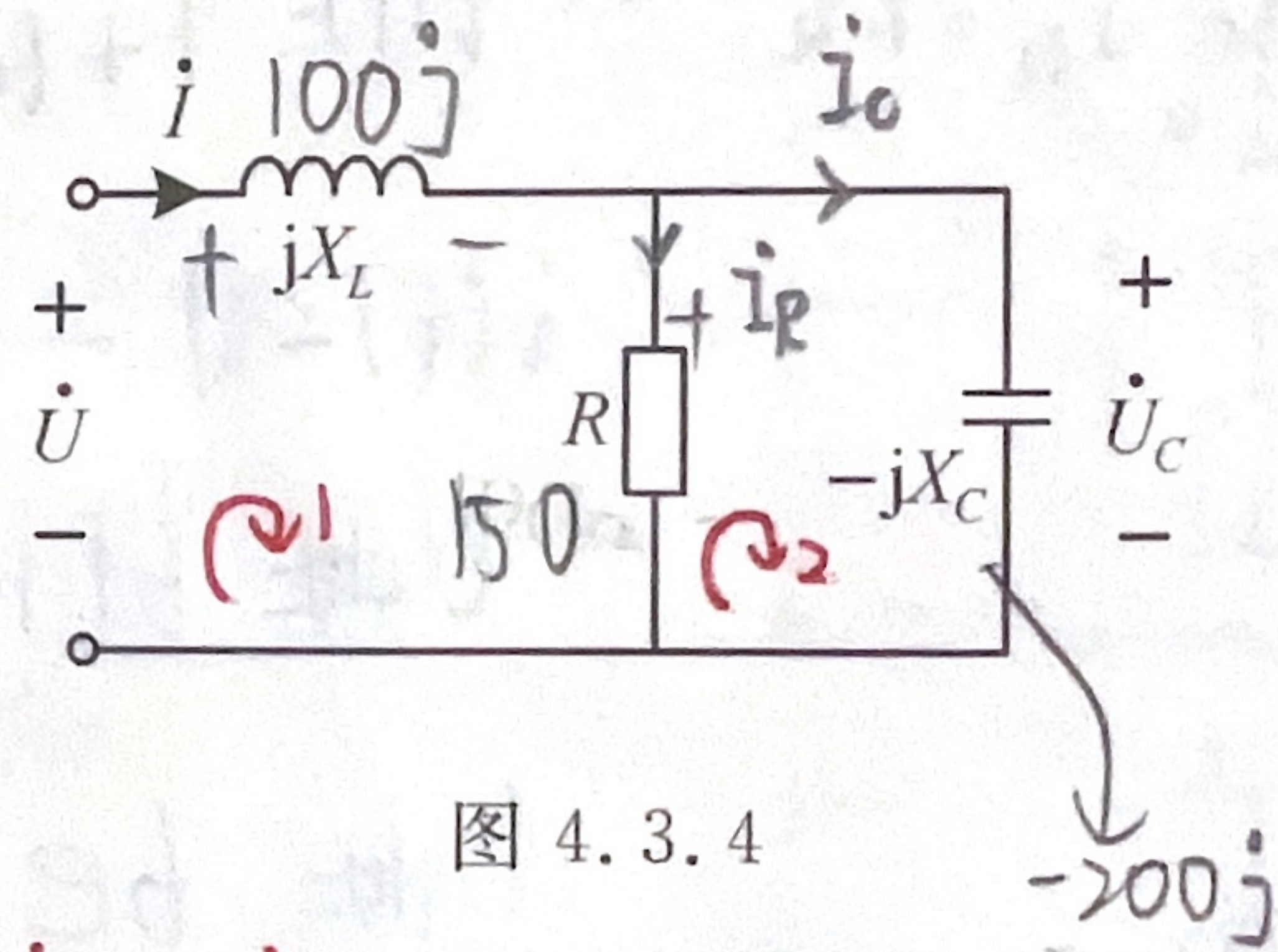
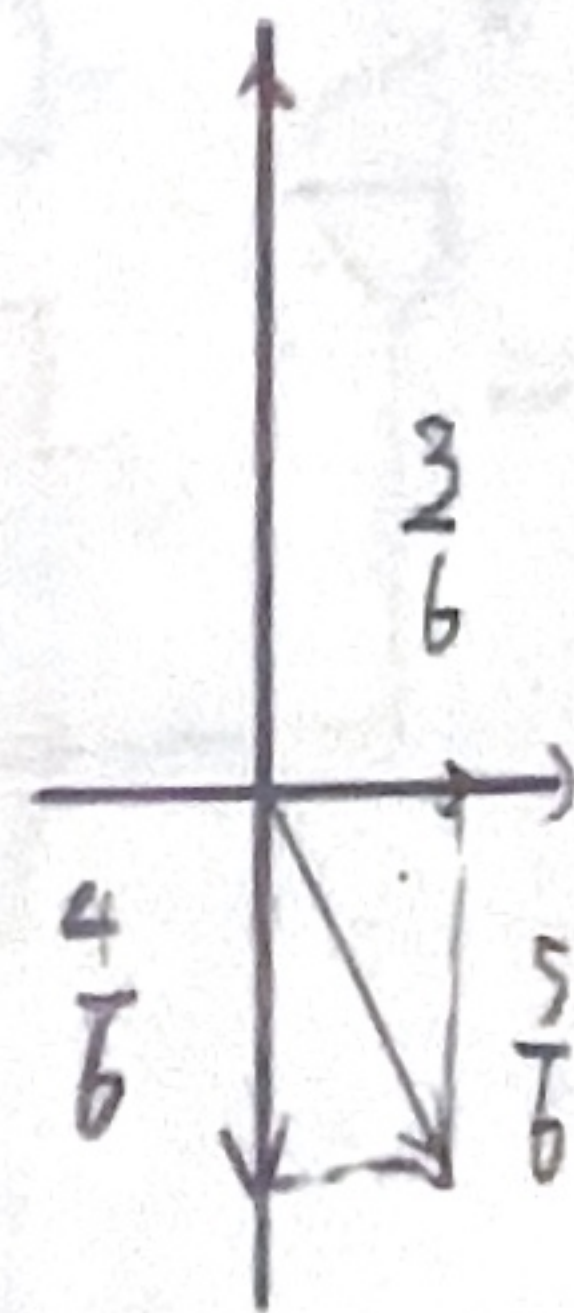


图 4.3.4



断路法求 U_{oc}

$$U_a - U_b = \frac{6}{6 + j10 + 6} \cdot 6 + \frac{6}{12 + j10} \times 5j$$

网孔法求 I_{sc}

$$\begin{cases} (12 + j10)i_1 - 6i_2 = 6 \\ -6i_1 + 6i_2 = 5j \cdot i_1 \end{cases}$$

回路2 KVL: $150\dot{I}_R = \dot{I}_C(-200j) - 0$

电容元件: $-200j \cdot \dot{I}_C = \dot{U}_C$ 代数形式

代入①得 $\dot{I}_R = -2/3j \cdot 1\angle 0^\circ = \frac{2}{3}\angle 0^\circ + 270^\circ$
 $200\dot{I}_C = 100 \Rightarrow \dot{I}_C = \frac{1}{2}A$
 $\Rightarrow I = \dot{I}_R + \dot{I}_C = \frac{2}{3}\angle 0^\circ + 270^\circ + \frac{1}{2}\angle 0^\circ$